

Phase 5: Bill of Materials

Marklin WiFi AC Train Controller

2026-03-04

Test Project: Bill of Materials

BOM Statistics

Metric	Count
Unique part numbers	24
Total component count	43
Active ICs/modules	8
Discrete	7
semiconductors Passives (R/C)	25
Connectors/switches	5
Estimated BOM cost (1x)	~\$3.90-4.10

KiCad Library Status

Status	Cou	Components
Available in KiCad	19	MB6S, XL1509, SS34, BAT54S, 1N4148WS, BT136-600, MOC3021M, USBLC6-2SC6, all passives, LED, connectors,
Substitute symbol available	1	switches SN74LVC00A (use 74HC00 symbol, set value to 74LVC00A)
Import needed (easyeda2kicad)	2	ESP32-C3-MINI-1-N4, EL357N
Footprint check needed	2	MOC3021S-TA1 (SMD-6P variant), snubber caps (400V film)

Components Requiring Import

These components are not in the KiCad standard libraries and must be imported before schematic capture:

1. ESP32-C3-MINI-1-N4 (U1)

- LCSC: C2838502
- Import command: easyeda2kicad --full --lcsc_id C2838502
- Reason: KiCad has ESP32-C3-WROOM-02 and DevKitM-1, but not the MINI-1 module
- Priority: High - central MCU module

2. EL357N(C)(TA)-G (U5, U6)

- LCSC: C29981
- Import command: easyeda2kicad --full --lcsc_id C29981
- Reason: Not in Isolator library. PC817 has same pinout but EL357N is SOP-4 (SMD) with different footprint
- Priority: Medium - can alternatively use PC817 symbol with custom footprint
- Qty: 2 (zero-crossing detector + voltage sense)

Substitute Symbols

SN74LVC00APWR (U3)

- Use KiCad symbol 74xx:74HC00 - identical logic function and pinout
- Set component Value field to SN74LVC00A
- Assign footprint Package_SO:TSSOP-14_4.4x5mm_P0.65mm
- No import needed

BOM CSV

The full BOM is exported to workflow/05-bom.csv with the following columns:

- Item, Block, Designator, Description, Manufacturer, MPN, Package, LCSC, Qty, Unit Price, KiCad Symbol, KiCad Footprint, Status

The CSV is suitable for import in Excel or LibreOffice Calc.